

Automated Satellite-AI Detection of Unstable Ice Patches for Cryo-Hydrological Hazard Early Warning in the Himalaya

Project documentation / research concept note **Domain:** Remote Sensing · Geospatial Machine Learning · Natural Hazards **Region:** Central Himalaya (India) — pilot basin: Bhagirathi/Srikanta around Dharali, Uttarkashi, Uttarakhand **Level:** Undergraduate research internship → conference/journal paper

1. One-line thesis

Existing research is **forensic** — it explains past ice-patch collapses *after* they occur. This project develops the **first automated satellite-AI pipeline** to (a) **detect** exposed, unstable ice patches in deglaciating nivation zones and (b) **map collapse-susceptibility** across a Himalayan basin, turning ISRO's identified early-warning signal into an operational, scalable screening system — *before* disasters occur.

2. Problem statement

Climate-driven deglaciation is exposing unstable patches of bare glacier ice on steep high-altitude slopes (nivation zones). When such a patch collapses, it releases a high-speed surge of ice, meltwater, and debris — a flash flood with **no warning**.

Critically, this hazard is **invisible to existing early-warning systems**: - It is **not a cloudburst** (no triggering rainfall). - It is **not a classic GLOF** (Glacial Lake Outburst Flood) — there is **no growing lake** to monitor as a precursor.

The 5 August 2025 **Dharali (Uttarkashi) flash flood** was traced by ISRO to exactly this mechanism: collapse of an exposed ice patch in the nivation zone of the **Srikanta Glacier**. The ISRO study (*npj Natural Hazards*, 2026) reconstructed the event after it happened and concluded that **exposed ice on steep slopes is an early-warning signal** — but stated the precursors "remain poorly constrained" and built **no predictive or automated detection system**.

The gap this project fills: there is currently no automated, scalable method to *find* such unstable ice patches across a region and *flag* the locations most likely to fail. That is the contribution.

3. Background and motivation (why now, why this is novel)

Fact	Source
Dharali flash flood (Aug 2025) caused by ice-patch collapse, not cloudburst/GLOF	ISRO, <i>npj Natural Hazards</i> 2026

Fact	Source
Cryo-hydrological hazards under deglaciation are "emerging and under-recognized"	same
India has 7,500+ glacial lakes; ~2 million people downstream at risk	NDMA / Eos
GLOF (lake-based) prediction is heavily studied; ice-patch collapse has ~1 paper	literature scan

Novelty position: - Lake-based GLOF prediction → **saturated** (many ML papers: logistic regression, ANN, Bayesian NN, Fuzzy-AHP). - Ice-patch / nivation-zone collapse → **brand-new hazard class (named 2026), one forensic paper, zero predictive/automated work.**

This is the right kind of gap for a publishable undergraduate paper: real-world urgency, a clearly identified opening, and a tractable scope.

4. Scope and claims (calibrated to evidence — read this carefully)

The reliability of the paper depends on claiming **only what the data can support**.

What we claim (defensible): 1. **Detection** — automatically locate exposed, unstable ice patches from satellite + terrain data. *Measurable with standard accuracy metrics.* 2. **Susceptibility mapping** — rank slope-units by how closely their conditions match those that preceded Dharali. *Validated as a case study + condition analysis.* 3. **Automation & scalability** — do across a whole basin what ISRO did by hand for one slope.

What we do NOT claim (would get rejected): - ✗ Predicting the *exact time* a patch will collapse (only ~1 labeled event exists — impossible to validate). - ✗ A live binary "collapse alarm."

Framing: this is a **screening / prioritization tool** ("monitor these N slopes closely"), analogous to landslide-susceptibility or seismic-hazard zoning — which reliably tell you *where to worry*, not *when* the event hits. Real-time timing = explicitly **future work**.

5. Objectives

1. Build an automated pipeline to detect exposed bare-ice patches in a Himalayan basin from open satellite data.
2. Derive terrain and deglaciation precursor features (slope, aspect, elevation, multi-year ice-area change).
3. Produce a **collapse-susceptibility map** combining ice detection + precursor features.
4. Validate against the Dharali/Srikanta event and known-stable control sites.
5. Quantify accuracy and explicitly characterize limitations.

6. Datasets (all free / open)

Dataset	Role	Resolution	Access
Sentinel-2 MSI	Ice/snow/rock classification (NDSI, SWIR bands)	10–20 m, 5-day	Copernicus / Google Earth Engine (GEE)

Dataset	Role	Resolution	Access
Sentinel-1 SAR	All-weather surface change, cloud penetration	10 m	Copernicus / GEE
Copernicus DEM (GLO-30)	Slope, aspect, elevation	30 m	OpenTopography / GEE
ALOS PALSAR / HMA 8 m DEM	Higher-res terrain (optional)	8–12 m	OpenTopography / NSIDC
Randolph Glacier Inventory (RGI 7.0)	Glacier outlines (define study area)	vector	GLIMS / NSIDC
Hi-MAG / HMA glacial-lake inventory	Distinguish lake vs non-lake hazard	30 m	Zenodo 10.5281/zenodo.4275164
ERA5-Land	Temperature / melt forcing	~9 km	Copernicus CDS
News + public video/photo archives	Event labels (Dharali; other reported sites)	—	manual compilation

Recommended platform: **Google Earth Engine** (free for research) — handles Sentinel + DEM at scale without downloading terabytes.

7. Step-by-step model approach

Stage 0 — Study area & data preparation

- Define the pilot basin (Srikanta / Bhagirathi headwaters near Dharali) using RGI glacier outlines + a watershed boundary.
- Assemble cloud-free Sentinel-2 composites for the **ablation season (Jun–Sep)** across multiple years (e.g., 2017–2025).
- Co-register all layers to a common grid (e.g., 10 m).

Stage 1 — Surface classification (detect bare ice)

- Compute spectral indices: **NDSI** (snow/ice), **NDWI**, SWIR ratios to separate **bare ice vs snow/firn vs rock vs debris**.
- Train a pixel classifier (**Random Forest** is the reliable baseline; optionally a small **U-Net / CNN** for spatial segmentation) on hand-labeled training polygons.
- Output: per-pixel class map → extract the **"exposed bare-ice" mask**.
- *Metric*: overall accuracy, per-class precision/recall, IoU on a held-out validation set.

Stage 2 — Precursor feature engineering

For each ice-bearing slope-unit, derive the conditions ISRO linked to collapse: - **Slope angle** (steep = unstable) — from DEM. - **Aspect** (N/NE-facing matched Dharali) — from DEM. - **Elevation** (nivation-zone altitude band) — from DEM. - **Multi-year exposed-ice-area change** (thinning/shrinking trend) — Stage-1 mask differenced across years. - **Distance to/absence of a glacial lake** (confirms non-GLOF mechanism) — from

Hi-MAG. - **Melt-season temperature** — ERA5-Land. - (*optional*) SAR coherence loss = surface disturbance — Sentinel-1.

Stage 3 — Susceptibility scoring

Two routes, pick per ambition: - **3a. Knowledge-driven (robust, interpretable)**: weighted index of the precursor features (e.g., **AHP / Fuzzy-AHP**, the accepted method in GLOF literature). Produces a transparent 0–1 susceptibility score with no need for many training events. - **3b. Data-driven (higher novelty)**: train a classifier (**Random Forest / XGBoost**) where the positive class = the Dharali precursor conditions and confirmed-stable slopes = negatives; use **probability output as susceptibility**. Use **feature importance / SHAP** to show *which* conditions drive risk. - Output: a basin-wide **collapse-susceptibility map** (low → high).

Stage 4 — Validation

- **Positive check**: does the Srikanta/Dharali ice patch fall in the **high-susceptibility** class in *pre-event* imagery? (the key result)
- **Negative/control check**: stable glaciated slopes with exposed ice that did *not* collapse should score lower → demonstrates discrimination, not just "all ice = high."
- **Temporal check**: run the model on 2017–2024 imagery; confirm the Dharali patch trended toward higher susceptibility before Aug 2025.
- **Sensitivity analysis**: vary feature weights/thresholds; report how stable the ranking is.
- **Metrics**: classification accuracy (Stage 1); ROC-AUC / success-rate curve for the susceptibility map (standard in landslide-susceptibility validation); qualitative case-study confirmation.

Stage 5 — Output product & interpretation

- Deliver a **prioritization map**: "these N slope-units carry collapse-prone conditions — monitor closely."
- Discuss operational use (where authorities should focus ground/drone monitoring).
- Clearly state what the map does **not** say (timing).

8. Reliability & honest limitations (put this in the paper)

Limitation	Mitigation / framing
Very few labeled collapse events (~1: Dharali)	Use <i>susceptibility</i> (condition-matching), not event-forecasting. Knowledge-driven scoring (3a) needs no large positive set.
False positives (ice ≠ guaranteed collapse)	Frame as screening/prioritization, not a binary alarm.
No collapse-timing capability	Explicitly future work.
Cloud cover / seasonal snow masks ice	Use ablation-season composites + Sentinel-1 SAR as backup.
DEM resolution limits steep micro-terrain	Use highest-res DEM available (ALOS/HMA 8 m); note as constraint.
Single-basin pilot	Frame as proof-of-concept; multi-basin = future work.

Reliability summary: reliable as a **detection + screening** system with quantified accuracy; *not* a deterministic collapse predictor — and the paper succeeds precisely because it does not overclaim.

9. Expected outcomes / contributions

1. First **automated** detector of exposed unstable ice patches for this hazard class.
 2. First basin-scale **collapse-susceptibility map** for nivation-zone ice patches.
 3. Operationalization of ISRO's qualitative "early-warning signal" into a quantitative, scalable product.
 4. An open, reproducible GEE/Python workflow others can extend to new basins.
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10. Suggested tech stack

- **Google Earth Engine** (Python API / `geemap`) — data access + large-area processing
 - **Python:** `numpy`, `pandas`, `rasterio`, `geopandas`, `scikit-learn`, `xgboost`
 - **Deep learning (optional):** `PyTorch` / `TensorFlow` for U-Net segmentation
 - **Explainability:** `SHAP`
 - **Visualization:** `matplotlib`, `QGIS` for final maps
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11. Indicative timeline (8–12 week internship)

Weeks	Work
1–2	Literature lock-in, study-area + data setup in GEE
3–4	Stage 1: ice/snow/rock classification + validation
5–6	Stage 2–3: precursor features + susceptibility model
7–8	Stage 4: validation against Dharali + controls
9–10	Maps, figures, sensitivity analysis
11–12	Paper writing + submission draft

12. Candidate publication venues

- *npj Natural Hazards*, *Natural Hazards* (Springer), *Remote Sensing* (MDPI), *Journal of Mountain Science*
 - Conferences: IGARSS, ISRS (Indian Society of Remote Sensing), regional climate/hazard symposia
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13. Key references

1. ISRO et al. (2026). *Ice-patch collapse and early-warning implications from a Himalayan flash flood: emerging cryo-hydrological hazards under deglaciation*. *npj Natural Hazards*. <https://www.nature.com/articles/s44304-026-00191-x>

2. The Tribune (2026). *ISRO study identifies ice-patch collapse as cause of August 2025 Uttarkashi flash flood*.
 3. ML-based probabilistic prediction of glacial lake formation (2025). Scientific Reports. <https://www.nature.com/articles/s41598-025-17401-7>
 4. Fuzzy-AHP susceptibility assessment of GLOFs in the Indian Himalaya (2026). Applied Geomatics. <https://link.springer.com/article/10.1007/s12518-026-00698-y>
 5. Annual 30 m dataset for glacial lakes in HMA 2008–2017 (Hi-MAG). ESSD. <https://essd.copernicus.org/articles/13/741/2021/>
 6. Automated satellite-based glacial lake inventory and change detection in HMA (2026). Scientific Reports. <https://www.nature.com/articles/s41598-026-35446-0>
 7. A Monitoring Network for Mitigating Himalayan GLOFs (2025). BAMS.
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Prepared as an internship research concept note. Claims are deliberately calibrated to available evidence: the system performs detection and susceptibility screening, not deterministic event-time prediction.